

HayrideTM

GreenEarth **PeopleSafe**

 Proudly Indian **Green Cleaning** Technology

Food & Industry Hygiene & Disinfection Range



GreenPro Certified



By CII-IGBC

India's
First
Cleaning
Chemicals

FDA Licensed, GMP



MP FDA & GMP Certificate



Following International Safety Norms of



World Health
Organization



EU Parliament



US FDA GRAS: Food Safety

Proudly Indian, Technology 
Driven & Ethical Small Business



GreenEarth **PeopleSafe**

 TM

Clean & Disinfect with Highest Safety Standards

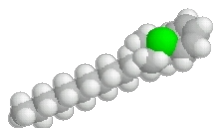


Following US FDA GRAS - Generally Regarded as Safe for Food Contact & US EPA Standards for all Ingredients used in Pharmaceutical Hygiene Range.

US FDA GRAS: Food Safety

Disinfectant Molecules -

(GRAS | US EPA | WHO)



Benzalkonium Chloride

Biocidal Activity:

Disruption of inter molecular interactions. This can cause dissociation of cellular membrane lipid bilayers, which compromises cellular permeability controls and induces leakage of cellular contents.

Efficacy:

Gram positive & negative bacteria including MRSA strains at 50 to 500 ppm.

Viruses such as H1N1, HIV, Hepatitis B & SARS at 1800-2000 ppm.

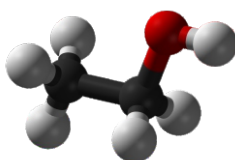
Fungi & Yeasts MIC is 2500 ppm.

Benzalkonium Chloride > 0.02% or 200 ppm effective against COVID 19 in for Hard Surface as per Research paper by published in the Journal of Hospital Infection.

[https://www.journalofhospitalinfection.com/article/S0195-6701\(20\)30547-8/fulltext#secsectitle0040](https://www.journalofhospitalinfection.com/article/S0195-6701(20)30547-8/fulltext#secsectitle0040)

Safety:

Used Widely in Ophthalmic Eye Solutions as preservative & skin Contact products like Medicated Strips for many years. It is even approved as a no Rinse Food Surface Cleaner at up to 250 ppm by US FDA. It is one of the safest anti microbial in current use in the world



Ethanol

Biocidal Activity:

Little is known about the specific mode of action of alcohols, but based on the increased efficacy in the presence of water, it is generally believed that they cause membrane damage and rapid denaturation of proteins, & subsequent interference with metabolism and cell lysis.

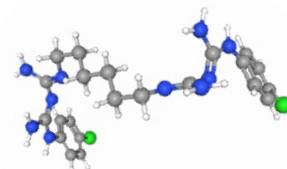
Efficacy:

Ethanol exhibits rapid broad-spectrum antimicrobial activity against vegetative bacteria & mycobacteria, viruses, and fungi. Ethanol is effective against almost all species at over 62% - 70% & above & is preferred in hospitals.

Safety:

Ethanol is Biodegradable Eco friendly naturally derived from grain & has very low inhalation & oral toxicity compared to synthetically produced highly toxic n-propanol. Ethanol is more potent than propanol on viruses even though propanol is slightly more potent on bacteria but since our products use a combination with Benzalkonium Chloride this is taken care of.

Ethanol above 62% active against Hep B & H1N1 as per Lab tests.



Chlorhexidine Gluconate

Biocidal Activity:

Chlorhexidine Gluconate is the gluconate salt form of chlorhexidine, a biguanide compound used as an antiseptic agent with topical antibacterial activity. Chlorhexidine gluconate is positively charged and reacts with the negatively charged microbial cell surface, thereby destroying the integrity of the cell membrane. Subsequently, chlorhexidine gluconate penetrates into the cell and causes leakage of intracellular components leading to cell death.

Efficacy:

Since gram positive bacteria are more negatively charged, they are more sensitive to this agent

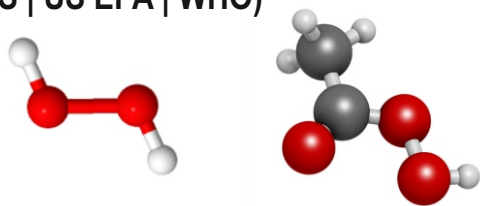
Safety:

Haylde Uses Chlorhexidine Gluconate in one of Its Hand Rubs. Chlorhexidine Gluconate is Human Safe & is commonly used in Mouth Washes as anti microbial agent.

Clean & Disinfect with Highest Safety Standards

Disinfectant Molecules -

(GRAS | US EPA | WHO)



Hydrogen Peroxide + Peracetic acid

Peracetic acid ($C_2H_4O_3$) is a mixture of acetic acid (CH_3COOH) and hydrogen peroxide (H_2O_2) in a watery solution. It is a bright, colorless liquid that has a piercing odor and a low pH value (2,8). Peracetic acid is produced by a reaction between hydrogen peroxide and acetic acid:

When peracetic acid dissolves in water, it disintegrates to hydrogen peroxide and acetic acid, which will fall apart to water, oxygen and carbon dioxide. Peracetic acid degradation products are non-toxic and can easily dissolve in water. Peracetic acid is a very powerful oxidant; the oxidation potential outranges that of chlorine and chlorine dioxide.

Uses

Peracetic acid is used mainly in the food industry, where it is applied as a cleanser and as a disinfectant. It can be applied during water purification as a disinfectant and for plumbing disinfection. Peracetic acid is suitable for cooling tower water disinfection; it effectively prevents bio film formation and controls Legionella bacteria.

Efficacy & Mechanism

Peracetic acid as a disinfectant oxidizes the outer cell membranes of microorganisms. The oxidation mechanism consists of electron transfer. When a stronger oxidant is used, the electrons are transferred to the microorganism much faster, causing the microorganism to be deactivated rapidly.

Peracetic acid can be applied for the deactivation of a large variety of pathogenic microorganisms. It also deactivates viruses and spores. Peracetic acid activity is hardly influenced by organic compounds that are present in the water. However, pH and temperature do influence peracetic acid activity. Peracetic acid is more effective when the pH value is 7 than at a pH range between 8 and 9. At a temperature of 15 °C and a pH value of 7, five times more peracetic acid is required to effectively deactivate pathogens than at a pH value of 7 and a temperature of 35 °C.

Active Surfactants -

(GRAS)

Naturally Derived Non Ionics

Alkyl Polyglucosides | Cocoamides

That are derived from Plant sources are completely & Readily Biodegradable. They are safe for Food Contact as Per US FDA Norms.

GRAS Approved Anionic

Sodium Lauryl Sulphate

Active Chelates -

(GRAS)

Naturally Derived Chelant

Sodium Gluconate

That are derived from Plant sources are completely & Readily Biodegradable. They are safe for Food Contact as Per US FDA Norms.



Safety Uncompromised -

All Products are free from all known Environmental & Human Toxic ingredients including -

APE/NPE | Phosphates | EDTA | Inorganic Acids | Syntheti Solvent Ethers etc.

General Factory Area (Non Active areas)



Stores
Packaging
Receiving
Dispatch
Utilities

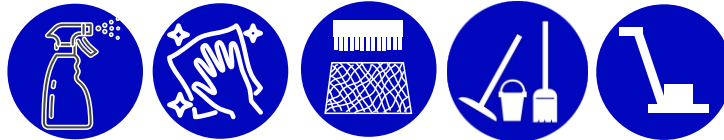
Cleaning Only



APC-F Green

Multipurpose Floor & Hard Surface Cleaner

GreenEarth
PeopleSafe™



Spray & Wipe (Hard Stains Scrub) | Mop, Machine Scrub

7ph Neutral
Mild Fragrance

Dilution: 1:100 to 1:50
or 1% or 10 ml/ltr to
2% or 20 ml/ltr



99% Readily Biodegradable &
many Natural Source Ingredients
**Free from NPE | Caustic | EDTA |
Phosphate | Synthetic Solvents**

Tested on 30 Eco & Safety Parameters

Certified by CII-Green Products & Services Council |
Approved for use in all IGBC certified Green Buildings



Safe for Aquatic Life
Passes 96 hr BioAssay Test



All Floors | Glass, Steel, Laminate | Furniture & Electronics

Cleaning & Disinfection



Sterimop MAX

Benzalkonium Chloride 3% w/v

Others: Alkyl Polyglucoside,
Denatured Ethanol, Sodium Gluconate
Fragrance, Approved Food Colour.

**MAX - Floor & Hard Surface Cleaner
& Disinfectant**



**96% Readily Biodegradable &
Natural Source Ingredients**

Free from NPE | Caustic | EDTA | Phosphate
No Synthetic Solvents | VOC < 0.5% in Dilution



MOPPING & SCRUBBING SPRAY & WIPE

7ph Neutral
All Surface Safe
Marble | Wood | Glass | Steel



Floor Mopping | Spray & Wipe

Dilution: 1:100 or 1% or 10 ml/ltr
300 ppm / 0.03% Benzalkonium Chloride

F&B Active areas - Non Food Contact Surfaces



Production Rooms
Mixing Rooms
Packaging Rooms

Cleaning & Disinfection



Sterishine



Benzalkonium Chloride 4% w/v

Others: Alkyl Polyglucoside, Denatured Ethanol,
Sodium Gluconate Approved Food Colour.

LFRA - Floor Cleaner & Disinfectant



Broad
Spectrum
Long Acting

Active
for up to
4 hours*

**Critical
Protection**

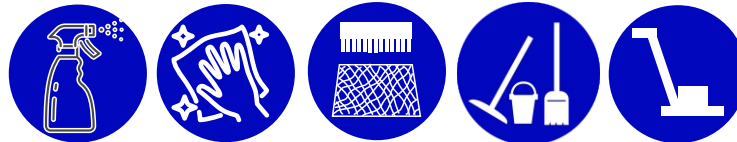
against HIV, #
H1N1 & Hep B
Mycobacteria
& Fungi

Dilution: 2-5%
or 20-50 ml/ltr
800-2000 ppm
Benzalkonium Chloride



OxyCleanz+

Multi-Surface Cleaner & Disinfectant



Spray & Wipe (Hard Stains Scrub) |

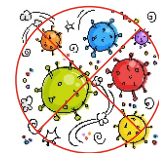
Mop, Machine Scrub



99% Readily Biodegradable &
many Natural Source Ingredients
**Free from NPE | Caustic | EDTA |
Chlorides | Synthetic Solvents**
Tested on 30 Eco & Safety Parameters

Certified by CII-Green Products & Services Council |
Approved for use in all IGBC certified Green Buildings

**GreenEarth
PeopleSafe™**



**Kills 99.99%*
Bacteria,
Fungi &
Microphages**

*Tested on 12 Microbes at 1:200 Dilution

**Super Concentrate
Use with Dilutor**

Dilution: 1:200 or
0.5% or 5 ml/ltr



All Floors | Glass, Steel, Laminate | Furniture & Electronics

F&B Active areas - Food Contact Surfaces (Rinse Off)

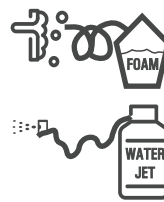


Production Vessels Pipelines

Cleaning only - Alkaline (Foaming)

UTILEX - CIP

Alkaline CIP/Manual Cleaner
Alkali Metal & Silica Salts 30-40% | Natural
source Food Safe Surfactant & Chelates



Phosphate Free

**US FDA GRAS
Listed Ingredients**

Usage:

CIP - 1st Step (Hot Water > 50C)
Dilution 1:33 or 30ml/ltr Water or 3%
(Contact/Recirculation Time - 3-5 minutes)
Rinse Water Recirculation Time - till pH 7

Also available

UTILEX - CIP LF

Low Foam Alkaline CIP Cleaner
For Fast Rinse Low Water consumption CIP



Gloves | Goggles | Face shield

Corrosive

Manual Cleaning Open Vessels -
1st Step (Hot Water > 50C)
Dilution 1:20 or 50ml/ltr
Water or 5% - Spray with Foam Generator
(Contact Time - 3-5 minutes) - Rinse with
Pressure Jet till pH 7



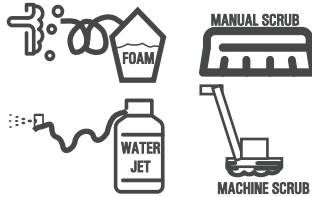
F&B Active areas - Food Contact Surfaces (Rinse Off)



Production Vessels
Conveyors
All Water Safe
hard Surfaces

Cleaning Machine & Manual - Neutral pH (Foaming)

UTILEX



Neutral Food Surface Detergent

Non Ionic & Anionic Surfactants | Natural

Source Chelates, Food Colour

Usage:

CIP - 1st Step (Hot Water > 50C)

Dilution 1:33 or 30ml/ltr Water or 3%

(Contact/Recirculation Time - 3-5 minutes)

Rinse Water Recirculation Time - 3-5 min

Manual Cleaning Open Vessels -

1st Step (Hot Water > 50C)

Dilution 1:20 or 50ml/ltr Water or 5% -

Manual Scrub/Rinse or Foam Generator spray

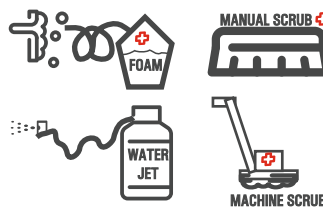
(Contact Time - 3-5 minutes) - Rinse with

Pressure Jet till no foam



Cleaning & Disinfection Neutral pH (Foaming)

Utilex-III



**Foaming Rinse off Food Contact Safe
Cleaner | Degreaser | Disinfectant**

Benzalkonium Chloride 3% w/v

Others: Alkyl Polyglucoside, Cocoamidopropyl
betaine, Sodium Gluconate, Food Color: Lemon Yellow.

Usage:

Manual Scrub & Rinse

CIP - High Temperature

Foam Generator - Foam & Rinse

Floor Surfaces - Scrub & Wipe

Dilution: - 1:20 to 1:50 2% to 5%

600 to 1500 ppm Benzalkonium Chloride



F&B Active areas - Food Contact Surfaces (Rinse Off)



Production Vessels
Pipelines

Disinfection CIP only

UTILEX Peroxy



Peroxy Acetic Acid ^{*}10-15%, Acetic Acid - 10-15%,
Hydrogen Peroxide 15-18%

US FDA GRAS
Listed Ingredients

Usage:

CIP only - after Cleaning

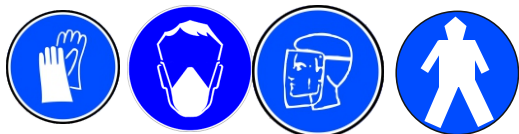
Dilution 1:100 or 10ml/ltr Water or 1%

PAA - 1000-1500 ppm or 0.1-0.15%

AA - 1000-1500 ppm or 0.1-0.15%

H2O2 - 1500-1800 ppm or 0.15-0.18%

(Contact Time - 1 - 5 minutes)



Gloves | Mask | Face shield | PPE Suit



Oxidizer



Corrosive



Danger
Hydrogen Peroxide

F&B Active areas - Food Contact Surfaces (Non Rinse)



Production Vessels
Conveyors
Pipelines
Counters

Disinfection only (Non Rinse application) - Dilutable

UtileX-SAN

Benzalkonium Chloride 2% w/v

Concentrated Disinfectant Solution

(No Rinse Food Safe Ingredients as per US FDA GRAS / US EPA @ 2% Dilution*)



US FDA GRAS
Listed Ingredients

Dilution: - 1% or 10 ml in 990 ml Water (200 ppm BKC)

Usage:



Hard Surfaces - Manual Mist Spray & Leave

CIP - Last Step

Floor Surfaces - Flood / Mist Spray

* Technical Document available separately with details of US FDA GRAS approval for No Rinse for food contact surfaces for Benzalkonium Chloride 200 ppm.



F&B Active areas - Non Food Contact Surfaces



Production Rooms
Mixing Rooms
Packaging Rooms

Fogging

Oxy-Fog



Hydrogen Peroxide 5% w/v

Others: Glycerine, Denatured Ethanol.

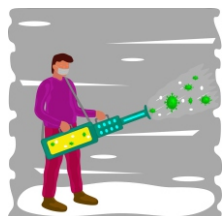
Dilution:

1:10 or 100 ml/ltr (5000 ppm) H2O2 - General

1:5 or 200 ml/ltr (10,000 ppm) H2O2 - Critical

DO NOT USE IN CONCENTRATE

Usage:



Fog area with Fogging Machine.
Ventilate area after 5 minutes.



Gloves | Mask | Face Shield | Irritant

Corrosive
in Concentrate

**US FDA GRAS
Listed Ingredients**



Quatfog



Disinfectant Concentrate - Spray/Fog

Dimethyl Benzyl ammonium Chloride 8% w/v |

Didecyl dimethyl ammonium chloride 8% w/v

Others: Glycerine, Denatured Ethanol.

Usage:



Wet area with ULV Sprayer / Fogging Machine.
Ventilate area after 10 minutes.



Gloves | Mask | Face Shield | Irritant

1:100 Dilution -
800 ppm DBAC
800 ppm DDAC



Hand Hygiene - all Staff (Unfragranced available)

Soap Free Hand Washes



Food Grade Un-fragranced Hand Wash
Made from Skin Friendly Laureth Sulfate.

Contains Aloe Vera & Glycerine.

Free From Slurry/Soap Oil (LABSA)

Excellent Foaming, Quick Rinsing,
Moisturized after Feel.

Alcohol Based Hand Sanitizers



SANITZ+

Disinfecting Hand Rub

Denatured Ethanol 70% v/v +
Benzalkonium Chloride 0.7% w/v



SANHEX

Disinfecting Hand Rub

Ethanol IP 70% v/v
Chlorhexidine Gluconate Solution IP 2.5% w/v
(Equivalent to Chlorhexidine Gluconate 0.5% w/v)



FDA Licensed

Efficacy Tested

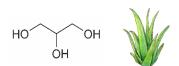
**Broad
Spectrum
Long Acting**

**Active
for up to
2-3 hours***

Low Toxicity

**Natural
Ethanol**
in place
of IPA

**- Comfort
Formula-**



**Moisturizing
Non Irritant**



Complete Eco Friendly Product Range

Manufacturing more than 90 speciality products for complete Hygiene needs of Housekeeping, Infection Control, Medical, Food-Pharma, Personal Care & Laundry.

World Class Manufacturing Facilities

ISO 9001:2008 Certified following WHO GMP Norms State-of-the-Art Manufacturing Facilities with Latest Machinery & Labs.

Top of the line Technical R&D

The group is driven by innovation since 1979. The R&D is headed by the promoters with over 40 years international Experience in Chemicals.

Beyond Products - Consultancy & Education

Three Tier Training & Education Program | Hygiene Audits | Cost Analysis | Awareness Posters | Sharing of World Best Practises

**Complete Housekeeping | Kitchen & Laundry Products available
- ask for separate Brochure**



Our Standard Setting Global Manufacturing Facility at Richhai Industrial Area Jabalpur - Commissioned September 2011

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